2.1 Writing and Graphing Inequalities

Name _____ Period ____ Date _____

Algebra I 2.1 · Student sheet

I can…

- write inequalities from words.
- check solutions of inequalities and graph them on a number line.
- write inequalities from graphs and real-life situations.

Vocabulary · fill in the missing word

Inequality — a statement that compares two _____ with <, >, ≤, or ≥

Solution of an inequality — a value that makes the inequality _____

Graph of an inequality — open circle for < or >, closed circle for ≤ or ≥, with the _____ shaded

Part 1 · Words to symbols

Phrases: at least · at most · more than · fewer than · no more than · no less than · below · minimum · maximum · exceeds

<	>	≤	≥

Write one real-life sentence for each symbol.

Part 2 · Notes: fill in the blanks

Open circle for _____ or _____; closed circle for _____ or _____.

“At least 5” means x ____ 5. “At most 5” means x ____ 5.

A solution of an inequality is a value that makes it _____.

Part 3 · Examples (work them with the class)

Example 1

Write each sentence as an inequality.

a. A number x plus 4 is less than 10.

b. Twice a number y is no more than −6.

Example 2

Is $x = -3$ a solution of $2x + 5 \geq -1$?

Example 3

Graph each inequality.

a. $x > -2$ b. $y \leq 4$

Example 4

You must be at least 16 years old to get a driver’s license. Write and graph an inequality for the ages a that qualify.

Part 4 · Partner practice

Partner 1

Write as an inequality: a number t minus 3 is more than 8.

Partner 2

A number line has a closed circle at 1 and is shaded to the left. Write the inequality.